Avery
Briggs

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Dear Hiring Team,

I am writing to express my interest in the Full Stack Developer position at BuildCode Inc. With a Bachelor of Computer Science from the University of New Brunswick and several years of experience building automation tools, internal software applications, and full-stack systems, I am excited about the opportunity to contribute to a team developing impactful technology for real-world applications.

In my previous role as a Process Analyst and Software Systems Developer at BWS Manufacturing, I designed and developed internal software applications used across multiple departments including Sales, Procurement, Production, and Scheduling. Much of my work focused on building Python-based applications, data pipelines, and interactive dashboards using tools such as Streamlit, SQL Server, and Pandas. These systems automated operational workflows, improved data visibility, and reduced manual processes across the organization.

Working within a small technical team required a high degree of autonomy and problem solving. I regularly analyzed operational challenges, designed technical solutions, and implemented applications that became part of daily workflows. Projects included automation tools, OCR-based data extraction systems, internal reporting dashboards, and visualization tools for warehouse inventory systems.

I am particularly interested in BuildCode Inc's mission to develop innovative technology solutions for enterprise and community partners. The opportunity to work on production systems while continuing to expand my experience with modern web frameworks and cloud platforms is very appealing to me.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my background in software development, automation systems, and data tools could contribute to your team.

Sincerely,
Avery Briggs

Professional Summary

Software developer with a Bachelor of Computer Science and experience building internal software systems, automation systems, and operational dashboards using Python and SQL. Experienced in developing data pipelines, automation utilities, and interactive dashboards used across multiple departments in a manufacturing environment. Strong background in problem solving, systems design, and rapid development of internal software applications.

Experience

Software Systems Developer | Process Analyst

BWS Manufacturing Ltd. | Centreville, N.B.

Jan 2021 – Feb 2026

Developed internal software systems and software development tools supporting operational workflows across Sales, Procurement, Production, and Scheduling within a small technical team in a manufacturing environment.

Key Contributions

- Full-stack development of Python-based internal applications and Streamlit dashboards used by multiple departments to analyze operational data and automate workflows and support decision-making.
- Developed complex SQL Server data pipelines and analytics queries enabling cross-department analysis of inventory, production, purchasing, and sales metrics.
- Built automation tools that eliminated manual operational tasks, including systems that automatically aggregate and prepare files for reporting and meetings, reducing manual preparation time from hours to seconds.
- Designed and implemented a warehouse visualization system mapping inventory locations onto spatial storage layouts, improving operational understanding of inventory distribution and supporting warehouse planning workflows.
- Developed tools using OCR techniques to extract structured data from legacy PDF documents, enabling automated reporting and analysis of previously inaccessible operational records.
- Refactored legacy Microsoft Access applications by consolidating duplicated VBA logic into reusable shared modules, eliminating 3–5 smaller databases and improving maintainability of the primary operational system.
- Built an administrative time-tracking interface integrating live ShopClock data, allowing supervisors to review, adjust, and approve employee shift records before payroll submission, improving visibility and coordination across multiple departments.
- Created and maintained reusable Python utility libraries supporting database access, file management, reporting, and UI helpers used across internal projects.
- Provided internal IT systems support and troubleshooting across hardware, networked devices, and software systems, working with external vendors for network infrastructure, ERP systems, cloud services, and building security.

Notable Projects:

- IT Request and Inventory Ticketing System. Built from scratch using SQL and Access.
- Custom drag-n-drop production schedule. CustomTkinter application packaged into an executable file.
- Streamlit dashboards for sales margins and inventory analysis including inputs and reporting.
- 3D warehouse visualization using Plotly, demonstrating inventory bottlenecks and improving transparency.
- PDF parsing of legacy reports. Performed using Python and modules: Pdfplumber, Streamlit, and Pytesseract.
- Implementation of Access version control protocols to maintain program integrity.
- Yellow Tag system to automatically flag, record, and report missed parts when picked for production.
- Diagnosed and repaired hardware issues in a K40 CO_2 laser engraver used for in-house VIN plate engraving.

Technologies used:

- Python, SQL Server, Streamlit, Pandas, Plotly, Pytesseract, CustomTkinter, Microsoft Access, VBA, Syspro8 ERP, ShopClock, NeWare Security

Other Job Experience:

- Basketball Referee – CBOC – Fredericton, NB – Oct 2013 – Mar 2020
- Kitchen Staff – FMI – Fredericton, NB, -- May 2015 – Sept 2017
- Farm Hand – Hunter Brother's Farm Market – Florenceville, NB, -- Jun 2011 – Aug 2013

Education

Bachelor of Computer Science

UNB – Fredericton N.B.

MAY 2020

- Specialization: Software Systems.
- Recipient of Governor Thomas Carleton Entry Scholarship.

Skills

- **Programming**: Python, SQL, VBA, Java
- **Frameworks / Libraries**: Streamlit, Pandas, Plotly, Tkinter, CustomTkinter, Pdfplumber, Pytesseract
- **Databases**: Microsoft SQL Server, Microsoft Access
- **APIs**: NHL Stats API, IMDB API, Fitbit API
- **Tools:** Git, Syspro ERP, ShopClock
- **Languages**: English (native), French (Intermediate oral proficiency)
- **Other**: Strong cross-team communication and collaboration. Self-directed problem solving and rapid learning.

Awards & Activities

- Finalist (Top 12) - Kaspersky Academy Partner's Summit (KAPS) Lisbon, Portugal (December 2019).
- Certified intermediate plus in the French oral proficiency exam (June 2013).
- UNB dormitory Orientation Committee (September 2014) & Social Representative (September 2014 – April 2015).
- UNB Relay for Life steering committee logistics team (September 2013 – February 2014).

Hobbies

3D printing and hardware projects, sports analytics software, recreational athletics (cycling, kayaking, basketball officiating)

Projects

- Live NHL Scoreboard showcasing game scores and standings data, with an interactive playoff bracket selection. Leverages NHL Stats API to show real-time figues, summarized
- STL file generation for automating the creation of strings of text using custom fonts and other parameters. Uses a Solid2 to generate files, and Streamlit interface to model and tweak them.
- Minesweeper Android mobile application, built using Java and Android Studio.
- Image editing Android application taking advantage of the Gestures library to offer a unique interface for tuning photos.
- Traffic simulation engine using Pygame. Mimics some elements from Cities Skylines traffic patterns.
- COVID-19 interaction simulator built with JavaFX modeling contagion spread in configurable populations.

References available upon request